Supporting Information:

Nanoparticles that deliver RNA to bone marrow identified by *in vivo* directed evolution

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Description of Supporting Information: Literature Review of Delivery to Endothelial Cells, Structure of 7C1, siRNA Sequences, Representative FACS Gating, Representative Normalization, Barcode Sequences, Diagram of PCR Scheme, Dynamic Light Scattering, LNP Library Chemical Composition, Enrichment Diagram, Winning LNP Confirmation, Library Diameter, Consistency of Technical Replicates, Mouse Weights, LNP Stability, LNP Ionizability, LNP Endocytosis, LNP Biodistribution, sgRNA Sequence, HSPC Indels

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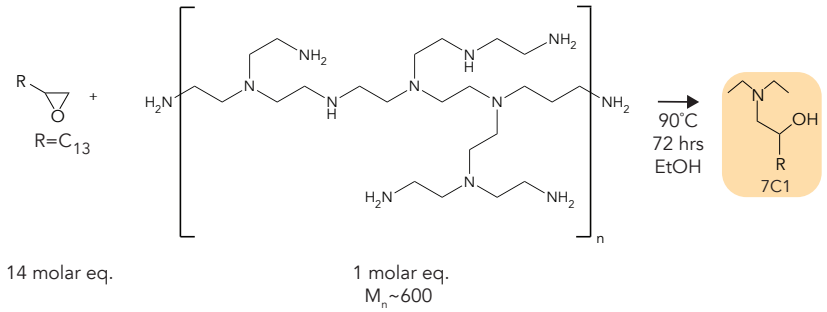
Figure S1. (A) Literature review of 8 papers reporting nanoparticles delivering siRNA to endothelial cells in various tissues *in vivo*. Efficacy is reported in either mRNA or protein silencing, as indicated. (B) Synthesis scheme for 7C1 lipomer. (C) Sequence and chemical modification patter for siICAM-2 and siLuciferase. (D) Representative FACS gating to isolate bone marrow endothelial cells. (E) Example of how normalized delivery is calculated. Barcode counts are normalized within a sample, and then to the DNA barcode input. This process is analogous to normalizing RNA-seq data. (F) Design of 91 nucleotide barcode used in this study, including 156 example 8 nucleotide DNA barcode sequences that are compatible on the same sequencing run. (G) Representation of two-step PCR scheme to amplify DNA barcodes and add identifiers for Illumina sequencing. (H) Diameters of 7C1 LNP formulated to either siRNA only or siRNA + DNA barcode. The siRNA: DNA barcode mass ratio is 10: 1.

Citation	Lung	Heart	Kidney	Target	Protein or mRNA
https://onlinelibrary.wiley.com/doi/epdf/10.1002/adhm.201400054	65% @ 1.25 mg / kg	-	-	Tie2	mRNA
https://onlinelibrary.wiley.com/doi/abs/10.1002/anie.201408221	35% @ 2.5 mg / kg	-	-	Tie2	mRNA
https://pubs.acs.org/doi/pdf/10.1021/nl5048972	95% @ 2.5 mg / kg	-	-	Tie2	mRNA
https://www.nature.com/articles/nnano.2014.84	90% @ 0.05 mg / kg	85% @ 0.6 mg / kg	80% @ 0.6 mg / kg	ICAM2	mRNA
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3982492/	80% @ 2.8 mg / kg	20% @ 2.8 mg / kg	25% @ 2.8 mg / kg	Tie2	mRNA
https://pubs.acs.org/doi/abs/10.1021/acsnano.8b03640	-	-	-	ICAM2	Pro
https://www.nature.com/articles/3302777	60% @ 1.9 mg / kg	30% @ 1.9 mg / kg	-	Tie2	mRNA
https://www.cell.com/action/showPdf?pii=S1525-0016%2816%2932706-X	-	-	-	OAT3	mRNA

Citation	Liver	Spleen	Pancreas	Target	Protein or mRNA
https://onlinelibrary.wiley.com/doi/epdf/10.1002/adhm.201400054	60% @ 1.25 mg / kg	50% @ 1.25 mg / kg	-	Tie2	mRNA
https://onlinelibrary.wiley.com/doi/abs/10.1002/anie.201408221	80% @ 1 mg / kg	25% @ 2.5 mg / kg	-	Tie2	mRNA
https://pubs.acs.org/doi/pdf/10.1021/nl5048972	-	85% @ 2.5 mg / kg	50% @ 2.5 mg / kg	Tie2	mRNA
https://www.nature.com/articles/nnano.2014.84	40% @ 0.15 mg / kg	-	-	ICAM2	mRNA
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3982492/	40% @ 2.8 mg / kg	-	-	Tie2	mRNA
https://pubs.acs.org/doi/abs/10.1021/acsnano.8b03640	75% @ 1.5 mg / kg	-	-	ICAM2	Pro
https://www.nature.com/articles/3302777	70% @ 1.9 mg / kg	-	-	Tie2	mRNA
https://www.cell.com/action/showPdf?pii=S1525-0016%2816%2932706-X	-	-	-	OAT3	mRNA

Citation	Lymph Node	Brain	Target	Protein or mRNA
https://onlinelibrary.wiley.com/doi/epdf/10.1002/adhm.201400054	-	-	Tie2	mRNA
https://onlinelibrary.wiley.com/doi/abs/10.1002/anie.201408221	-	-	Tie2	mRNA
https://pubs.acs.org/doi/pdf/10.1021/nl5048972	-	-	Tie2	mRNA
https://www.nature.com/articles/nnano.2014.84	40% (pro) @ 1.6 mg / kg	-	ICAM2	mRNA / Pro
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3982492/	-	-	Tie2	mRNA
https://pubs.acs.org/doi/abs/10.1021/acsnano.8b03640	-	-	ICAM2	Pro
https://www.nature.com/articles/3302777	-	-	Tie2	mRNA
https://www.cell.com/action/showPdf?pii=S1525-0016%2816%2932706-X	-	50% @ 30 mg / kg	OAT3	mRNA

b



14 molar eq. 1 molar eq. M_n~600

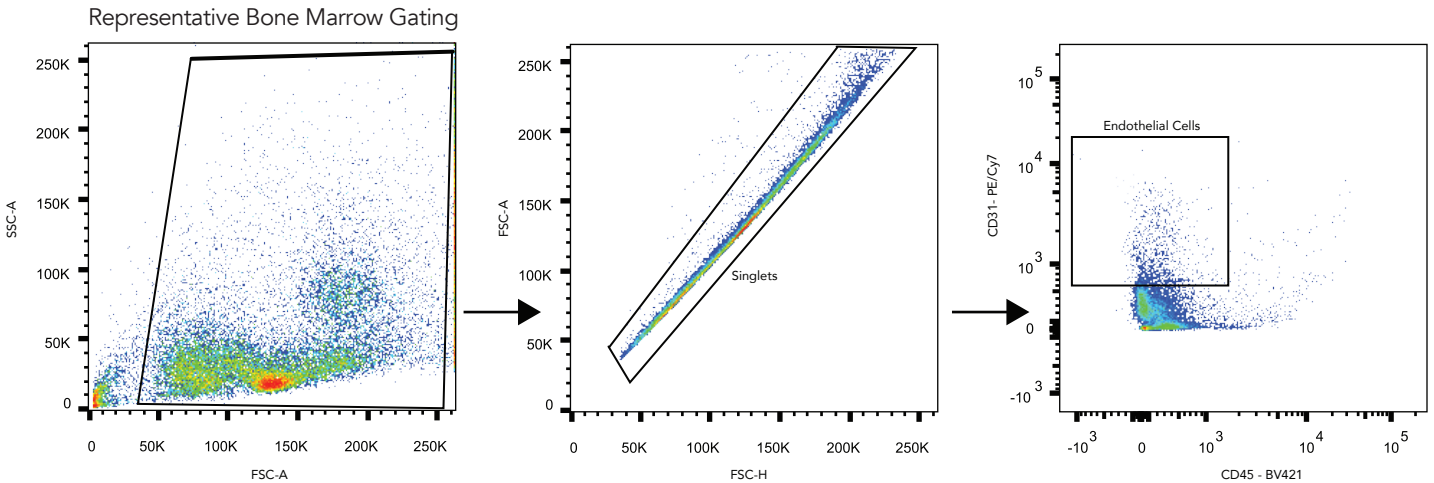
c

siICAM-2 sense: 5′ - AGGAcGGucucAAcuuuucdTsdT - 3′
siICAM-2 antisense: 5′ - GAAAAGUuGAGACCGUCCUdTsdT -3′

siLuciferase sense: 5′ - cuuAcGcuGAGuAcuucGAdTsdT- 3′
siLuciferase antisense: 5′ - UCGAAGuACUcAGCGuAAGdTsdT -3′

A, G, U, C: RNA nucleotide
dT: deoxy-T
a, g, u, c: 2′-O-Methyl nucleotide
s: phosphorothioate

d



E

		Raw Counts BMECs			
LNP		Mouse 1	Mouse 2	Mouse 3	Input
1	GACACAGT	200	80	200	100
2	GCATAACG	100	45	110	120
3	ACAGAGGT	240	105	250	110
Total Counts		540	230	560	330

		Normalized Counts BMECs (%)			
LNP		Mouse 1	Mouse 2	Mouse 3	Input
1	GACACAGT	37	34	36	30
2	GCATAACG	19	3	20	36
3	ACAGAGGT	44	46	45	33
Total (%)		100	100	100	100

		Normalized Counts BMECs (%)		
LNP		Mouse 1	Mouse 2	Mouse 3
1	GACACAGT	40	38	39
2	GCATAACG	17	18	18
3	ACAGAGGT	44	45	44
Total (%)		100	100	100

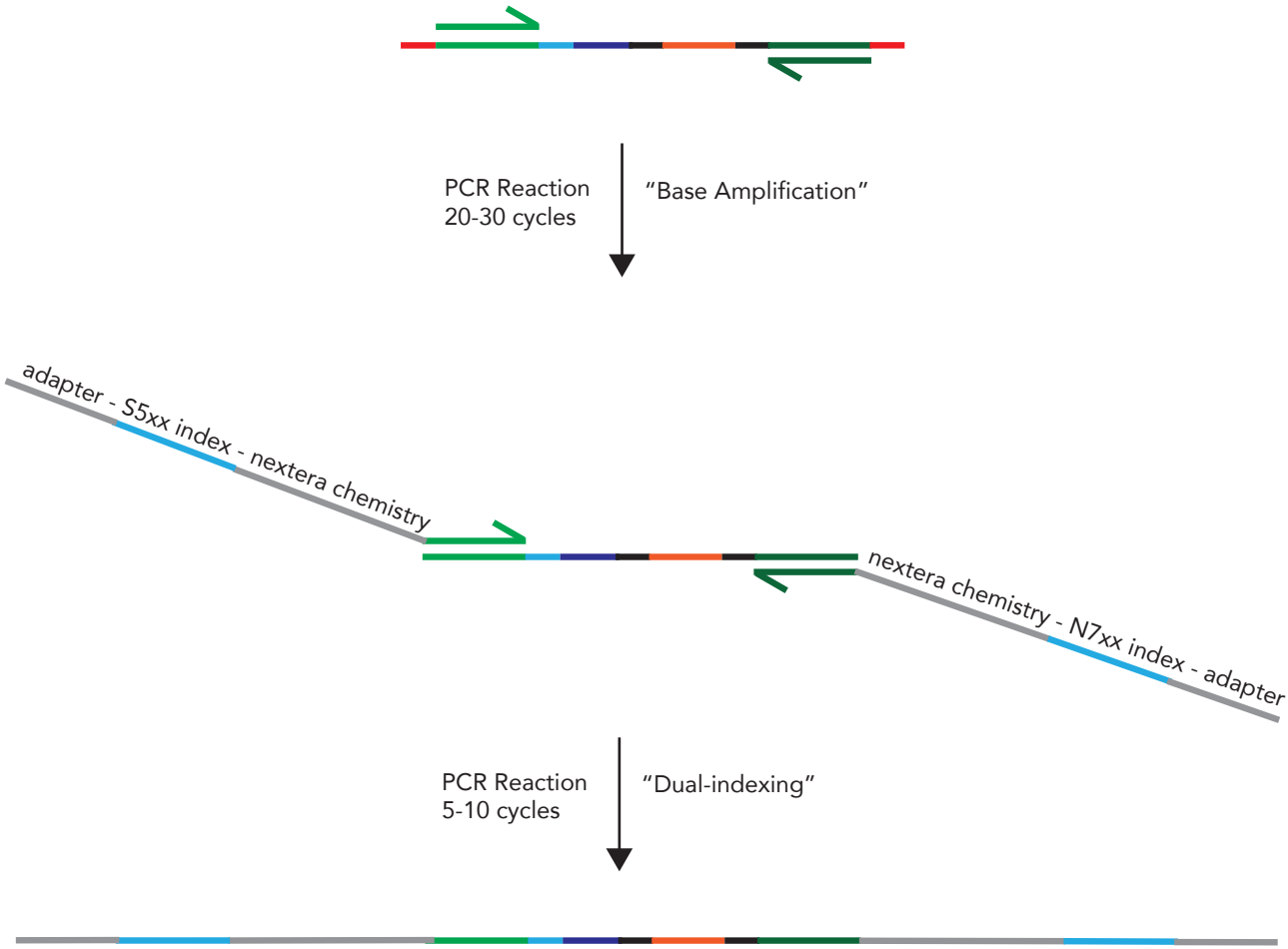
F

G*A*T*GCTCTCATACGAACTCGTCCNHNWCCTGCTAGTCCACGTCCATGTCCACCNWNH-8nt Barcode Seq-NWHGTGGTTAGTCGAGCAGAGAC*T*A*G

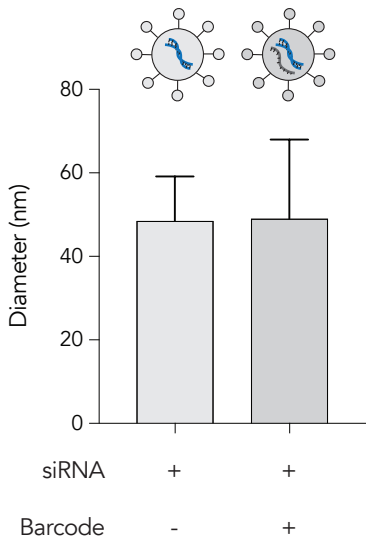
8nt Barcode Sequence

BC #	Seq	BC #	Seq	BC #	Seq	BC #	Seq
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2	GACGCAAT	41	TATGCCTT	80	GGTCGGTC	119	CTGACCGC
3	GCGAGTAT	42	GTAATTGC	81	TTGGATCC	120	TGACCAGG
4	ACCTAATC	43	GTCTCCGT	82	ATTGGTTC	121	CTCATAGG
5	AGGCGCTA	44	TGCATGGT	83	GATGGCCT	122	CCGTAAGC
6	GATCTACC	45	AGTCCGGT	84	TTATAGCA	123	CGAGACGT
7	CTACTGAT	46	TCCTGATG	85	GTCAATCT	124	GACGATAA
8	TGATCTAT	47	ATCGTCTA	86	CGCTCCGG	125	CCGCTGCT
9	ATGAGATG	48	GGACGTCC	87	ACTCAAGT	126	GGTTAGAA
10	GCGAATTC	49	CTACGAGG	88	CCGTTCGG	127	TTATCCGG
11	GATTCCGG	50	CAATCCGT	89	CCGCAGAG	128	AGTAGGTA
12	ATAATATA	51	GGCGCTTG	90	CGGTATCT	129	CGTACTAC
13	AGCATGCG	52	GTCCGTTA	91	TTATTAAT	130	AACTAGCG
14	GATTCAAC	53	GCCTCTCG	92	AGGCTCAT	131	TGCTCCTT
15	TACCTGCT	54	GAGAGTTG	93	TAGTACGT	132	TCGCCAAC
16	GCTAATCG	55	CATAATAG	94	AATATACG	133	CGCGGCTC
17	CTCCTTCG	56	TCTAGAGT	95	CGATGCTT	134	AAGGCGGT
18	ACGCTAGC	57	AAGTCTAG	96	CCAAGATT	135	GTAATGAG
19	GCAGGACT	58	ATTCGAGA	97	TCCATTAT	136	AGATACTA
20	ATTGCTCT	59	CTACCATT	98	AATACCAT	137	GAATCGTC
21	TACGCTCG	60	GTTAGTCA	99	CTGCGACC	138	AGGAAGAG
22	ACGCTCCA	61	ATAGAATC	100	GACTTGAG	139	CAGGTACC
23	CGGTCAAT	62	CTCAACTA	101	CAGAAGCA	140	TAGATAGC
24	CGCCTATT	63	CTTACGTC	102	TCTCCTAA	141	AGAGTAAG
25	TTGCGTTG	64	TGAGTTCG	103	CTGAGCCA	142	TCATTCCG
26	TCCTAAGA	65	ATGGTAGA	104	TCCTGCGC	143	CGGCGTCG
27	CAAGAAGG	66	TCCAGGCG	105	CGAACGCC	144	ATCAAGCA
28	TAGAATTA	67	CTCAGCAT	106	CTGCTCTA	145	TTGGCGTA
29	GGCGCCAA	68	TGCGTATA	107	GCCTACCA	146	CGTCCGCA
30	TAGATCCG	69	AATGCTAC	108	GGATGAAG	147	AGGACCGA
31	CGAGCAGC	70	CGCGAGGC	109	CTATATAC	148	CCTCGATC
32	TAAGATGA	71	GTCGAAGT	110	CGAATATG	149	TATCTGAG
33	AGCTCGGA	72	ACTATCTC	111	ACGCATTA	150	CGGAGTAA
34	TAACCGAA	73	GTCGCCTC	112	GGTAGACC	151	AGAATGAA
35	TATATCTA	74	AGTTACCG	113	CGTTATGC	152	AATCGGTT
36	AAGAGGAT	75	GAGTATAC	114	TCTGCGGA	153	CATCGCCA
37	ACGTCGAA	76	GGCAGTAG	115	CCTTGCAT	154	TATTGACT
38	CATCATTA	77	TGGAGACG	116	ATTATAGT	155	GTAGGCGG
39	TTGCAACT	78	ATTAGGAC	117	CTCGTAAT	156	GTTCGTAT

G


PCR Reaction
20-30 cycles
"Base Amplification"
adapter - S5xx index - nextera chemistry
nextera chemistry - N7xx index - adapter
PCR Reaction
5-10 cycles
"Dual-indexing"

H


Diameter (nm)
80
60
40
20
0
siRNA + +
Barcode - +

Supplementary Figure 2. (A) The chemical composition of 120 LNPs formulated in library 1. (B) Representative example of enrichment calculation. (C) Molar and chemical composition of top performing LNP in BMECs from Library 1. (D) Normalized ICAM-2 protein expression in BMECs 3 days after administration of the top performing LNP from screen 1 carrying either siLuc or siICAM2 at a dose of 1.0 mg / kg. (E) Comparison of potency in BMECs of original 7C1 compared to the top performing LNP from library 1. The top performing LNP from library 1 was ~2.2x more potent than 7C1. (F) The chemical composition of 40 LNPs formulated in library 2. (G) Hydrodynamic diameter of all 40 LNPs, as well as the control (the pooled LNPs). (H) Mouse characteristics (age, sex, weight) for all experiments. (I) Comparison of normalized DNA delivery in each individual technical replicate in library 1 by Paired One-way ANOVA. (J) Comparison of normalized DNA delivery in each individual technical replicate in library 2 by Paired One-way ANOVA. (K,L) Mouse weight over time for PBS- and library-treated mice for both library 1 and 2.

A

Library 1 Composition

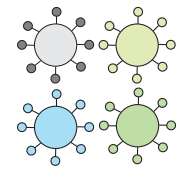
LNP #	7C1 Mole %	Chol %	PEG %	DSPC %	PEG Type		LNP #	7C1 Mole %	Chol %	PEG %	DSPC %	PEG Type
1	60	37.5	2.5	0	C14PEG2000		61	45	42.5	12.5	0	C18PEG2000
2	60	35	5	0	C14PEG2000		62	45	40	15	0	C18PEG2000
3	60	32.5	7.5	0	C14PEG2000		63	45	37.5	17.5	0	C18PEG2000
4	60	30	10	0	C14PEG2000		64	45	35	20	0	C18PEG2000
5	60	27.5	12.5	0	C14PEG2000		65	45	32.5	22.5	0	C18PEG2000
6	60	25	15	0	C14PEG2000		66	45	30	25	0	C18PEG2000
7	60	22.5	17.5	0	C14PEG2000		67	85	12.5	2.5	0	C18PEG2000
8	60	20	20	0	C14PEG2000		68	85	10	5	0	C18PEG2000
9	45	52.5	2.5	0	C14PEG2000		69	85	7.5	7.5	0	C18PEG2000
10	45	50	5	0	C14PEG2000		70	85	5	10	0	C18PEG2000
11	45	47.5	7.5	0	C14PEG2000		71	85	2.5	12.5	0	C18PEG2000
12	45	45	10	0	C14PEG2000		72	85	0	15	0	C18PEG2000
13	45	42.5	12.5	0	C14PEG2000		73	60	37.5	2.5	0	C14PEG3000
14	45	40	15	0	C14PEG2000		74	60	35	5	0	C14PEG3000
15	45	37.5	17.5	0	C14PEG2000		75	60	32.5	7.5	0	C14PEG3000
16	45	35	20	0	C14PEG2000		76	60	30	10	0	C14PEG3000
17	45	32.5	22.5	0	C14PEG2000		77	60	27.5	12.5	0	C14PEG3000
18	45	30	25	0	C14PEG2000		78	60	25	15	0	C14PEG3000
19	85	12.5	2.5	0	C14PEG2000		79	60	22.5	17.5	0	C14PEG3000
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21	85	7.5	7.5	0	C14PEG2000		81	45	52.5	2.5	0	C14PEG3000
22	85	5	10	0	C14PEG2000		82	45	50	5	0	C14PEG3000
23	85	2.5	12.5	0	C14PEG2000		83	45	47.5	7.5	0	C14PEG3000
24	85	0	15	0	C14PEG2000		84	45	45	10	0	C14PEG3000
25	60	37.5	2.5	0	C16PEG2000		85	45	42.5	12.5	0	C14PEG3000
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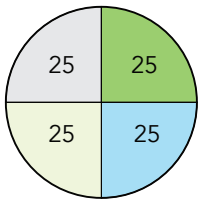
Library 1 Composition

LNP #	7C1 Mole %	Chol %	PEG %	DSPC %	PEG Type		LNP #	7C1 Mole %	Chol %	PEG %	DSPC %	PEG Type
1	60	37.5	2.5	0	C14PEG2000		61	45	42.5	12.5	0	C18PEG2000
2	60	35	5	0	C14PEG2000		62	45	40	15	0	C18PEG2000
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37	45	42.5	12.5	0	C16PEG2000		97	60	37.5	2.5	0	C18PEG3000
38	45	40	15	0	C16PEG2000		98	60	35	5	0	C18PEG3000
39	45	37.5	17.5	0	C16PEG2000		99	60	32.5	7.5	0	C18PEG3000
40	45	35	20	0	C16PEG2000		100	60	30	10	0	C18PEG3000
41	45	32.5	22.5	0	C16PEG2000		101	60	27.5	12.5	0	C18PEG3000
42	45	30	25	0	C16PEG2000		102	60	25	15	0	C18PEG3000
43	85	12.5	2.5	0	C16PEG2000		103	60	22.5	17.5	0	C18PEG3000
44	85	10	5	0	C16PEG2000		104	60	20	20	0	C18PEG3000
45	85	7.5	7.5	0	C16PEG2000		105	45	52.5	2.5	0	C18PEG3000
46	85	5	10	0	C16PEG2000		106	45	50	5	0	C18PEG3000
47	85	2.5	12.5	0	C16PEG2000		107	45	47.5	7.5	0	C18PEG3000
48	85	0	15	0	C16PEG2000		108	45	45	10	0	C18PEG3000
49	60	37.5	2.5	0	C18PEG2000		109	45	42.5	12.5	0	C18PEG3000
50	60	35	5	0	C18PEG2000		110	45	40	15	0	C18PEG3000
51	60	32.5	7.5	0	C18PEG2000		111	45	37.5	17.5	0	C18PEG3000
52	60	30	10	0	C18PEG2000		112	45	35	20	0	C18PEG3000
53	60	27.5	12.5	0	C18PEG2000		113	45	32.5	22.5	0	C18PEG3000
54	60	25	15	0	C18PEG2000		114	45	30	25	0	C18PEG3000
55	60	22.5	17.5	0	C18PEG2000		115	85	12.5	2.5	0	C18PEG3000
56	60	20	20	0	C18PEG2000		116	85	10	5	0	C18PEG3000
57	45	52.5	2.5	0	C18PEG2000		117	85	7.5	7.5	0	C18PEG3000
58	45	50	5	0	C18PEG2000		118	85	5	10	0	C18PEG3000
59	45	47.5	7.5	0	C18PEG2000		119	85	2.5	12.5	0	C18PEG3000
60	45	45	10	0	C18PEG2000		120	85	0	15	0	C18PEG3000

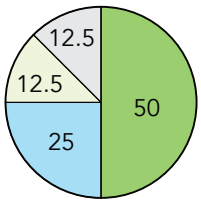
B



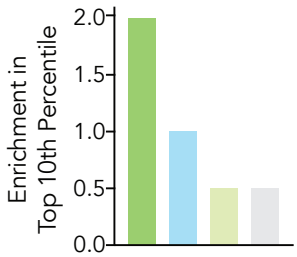
LNPs with 4
material properties



Initial
Library
(%)

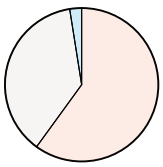


Top 10th
Percentile
(%)



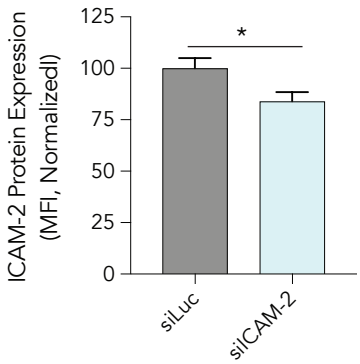
C

Screen 1 Winner

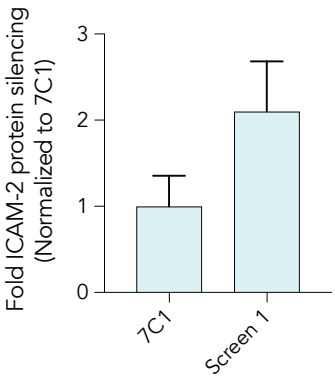


60% 7C1
37.5% Cholesterol
2.5% C_{16}PEG2000

D



E

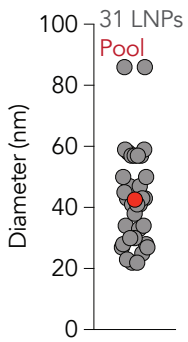


F

Library 2 Composition

LNP #	7C1 Mole %	Cholesterol Mole %	PEG Mole %	DSPC Mole %	
1	80	17.5	2.5	0	C16PEG2000
2	80	2.5	17.5	0	C16PEG2000
3	80	0	20	0	C16PEG2000
4	70	27.5	2.5	0	C16PEG2000
5	70	12.5	17.5	0	C16PEG2000
6	70	10	20	0	C16PEG2000
7	70	7.5	22.5	0	C16PEG2000
8	60	27.5	2.5	10	C16PEG2000
9	60	12.5	17.5	10	C16PEG2000
10	60	10	20	10	C16PEG2000
11	60	7.5	22.5	10	C16PEG2000
12	50	37.5	2.5	10	C16PEG2000
13	50	22.5	17.5	10	C16PEG2000
14	50	20	20	10	C16PEG2000
15	50	17.5	22.5	10	C16PEG2000
16	80	18.5	1.5	0	C16PEG2000
17	80	17.5	2.5	0	C18PEG2000
18	80	2.5	17.5	0	C18PEG2000
19	80	0	20	0	C18PEG2000
20	70	27.5	2.5	0	C18PEG2000
21	70	12.5	17.5	0	C18PEG2000
22	70	10	20	0	C18PEG2000
23	70	7.5	22.5	0	C18PEG2000
24	60	27.5	2.5	10	C18PEG2000
25	60	12.5	17.5	10	C18PEG2000
26	60	10	20	10	C18PEG2000
27	60	7.5	22.5	10	C18PEG2000
28	50	37.5	2.5	10	C18PEG2000
29	50	22.5	17.5	10	C18PEG2000
30	50	20	20	10	C18PEG2000
31	50	17.5	22.5	10	C18PEG2000
32	80	18.5	1.5	0	C18PEG2000
33	60	38.5	1.5	0	C16PEG2000
34	60	37.5	2.5	0	C16PEG2000
35	60	36.5	3.5	0	C16PEG2000
36	60	45	5	0	C16PEG2000
37	60	38.5	1.5	0	C18PEG2000
38	60	37.5	2.5	0	C18PEG2000
39	60	36.5	3.5	0	C18PEG2000
40	60	45	5	0	C18PEG2000

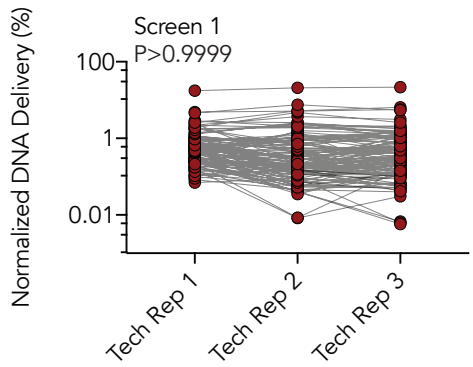
G



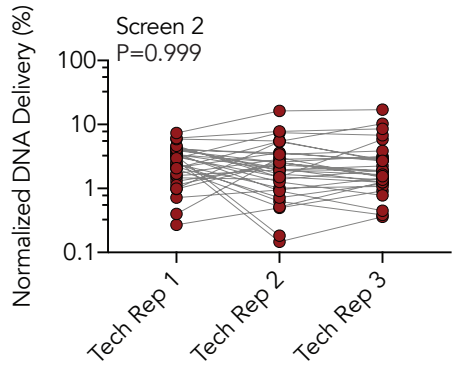
H

Experiment	Strain	Sex	Age	Weight (g)
Screen 1	C57BL/6J	F	7 weeks	19-21
Screen 2	C57BL/6J	F	5 weeks	15.2 - 18.3
Idv. LNP Testing	C57BL/6J	F	5 weeks	16.2 - 18.1
In Vivo Gene Editing	SpCas9 expressing on B6J background (Jax #026179)	F	5-7 weeks	17.0 - 18.4

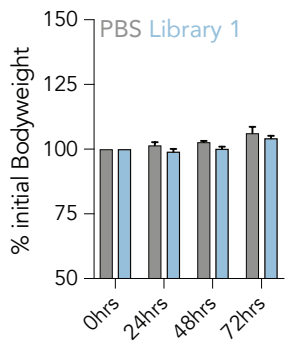
I



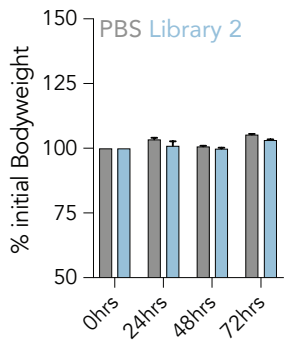
J



K

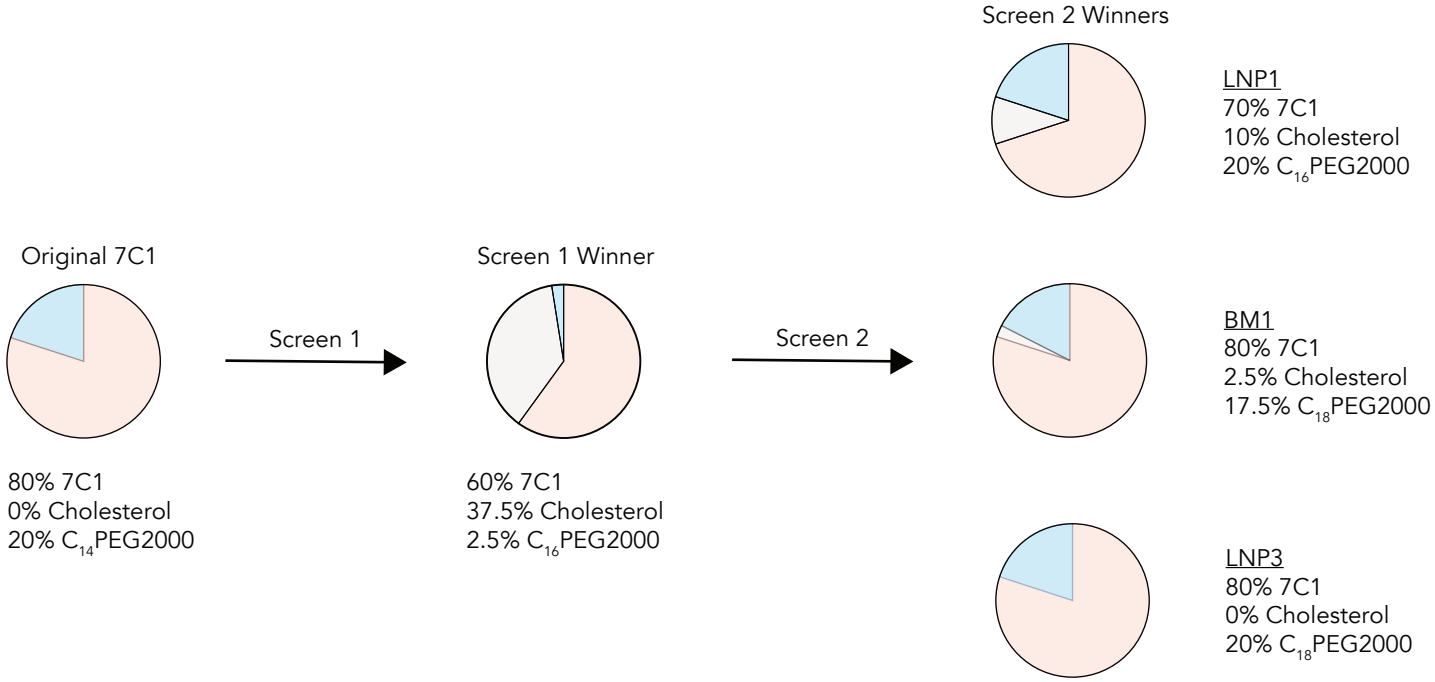


L

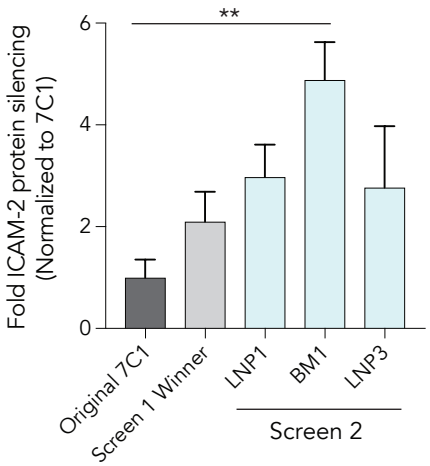


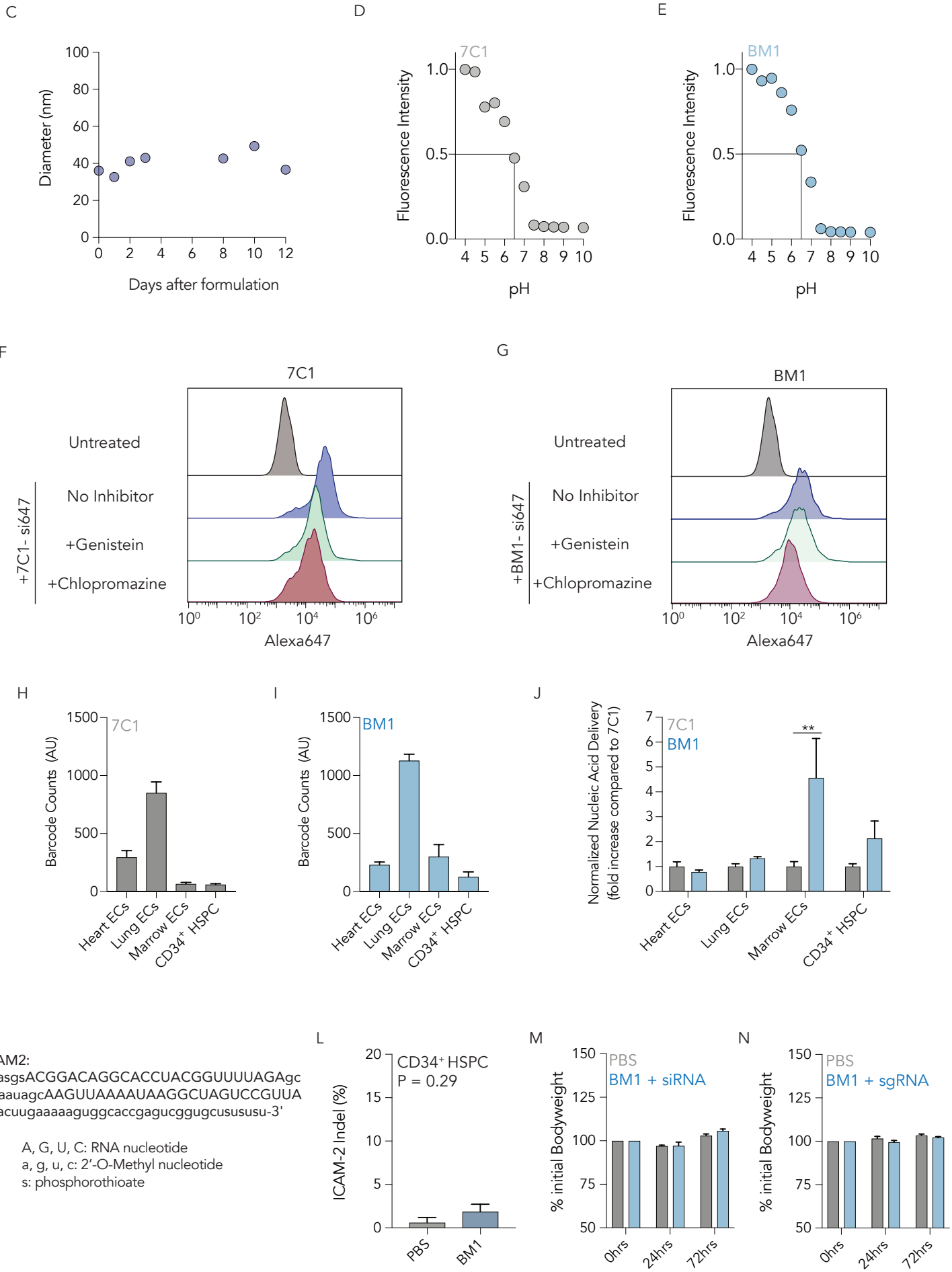
Supplementary Figure 3. (A) Molar and chemical compositions of 7C1, the top performing LNP from library 1, and the top 3 performing LNPs from library 3. (B) Potency of ICAM-2 silencing of 7C1, the top performing LNP from library 1, and the top 3 performing LNPs from library 3 in BMECs 3 days after injecting mice with a dose of 1.0 mg / kg siICAM-2. Notably, all top LNPs from library 2 were more potent than both 7C1 and the top performing LNP from library 1. (C) The top performing LNP from library 2 (BM1) is stable at 4°C for more than 10 days. (D) pKa of 7C1 LNP as measured by TNS assay. (E) pKa of BM1 LNP as measured by TNS assay. (F) Flow cytometry histogram of *in vitro* endocytosis of 7C1 carrying siRNA tagged with AlexaFluor647 in the presence of no endocytosis inhibitor, genistein (caveolin inhibitor), or chlorpromazine (clathrin inhibitor). (G) Flow cytometry histogram of *in vitro* endocytosis of BM1 carrying siRNA tagged with AlexaFluor647 in the presence of no endocytosis inhibitor, genistein (caveolin inhibitor), or chlorpromazine (clathrin inhibitor). (H) Biodistribution of 7C1 to endothelial cells in the heart, lung, and bone marrow, as well as $CD34^+$ HSPCs. (I) Biodistribution of BM1 to endothelial cells in the heart, lung, and bone marrow, as well as $CD34^+$ HSPCs. (J) Comparison of biodistribution between 7C1 and BM1. BM1 displays 4.6-fold higher biodistribution to BMECs than 7C1. Two-way ANOVA **P<0.01. (K) Sequence and chemical modification on sgICAM2. (L) Indels in $CD34^+$ HSPCs mediated by a 1 mg / kg injection of BM1 carrying sgICAM2. (M) Normalized weight of mice injected with 1.0 mg / kg BM1-siRNA compared to mice injected with PBS. (N) Normalized weight of mice injected with 1.0 mg / kg BM1-sgRNA compared to mice injected with PBS.

A


Original 7C1
80% 7C1
0% Cholesterol
20% C14PEG2000
Screen 1
Screen 1 Winner
60% 7C1
37.5% Cholesterol
2.5% C16PEG2000
Screen 2
Screen 2 Winners
LNP1
70% 7C1
10% Cholesterol
20% C16PEG2000
BM1
80% 7C1
2.5% Cholesterol
17.5% C18PEG2000
LNP3
80% 7C1
0% Cholesterol
20% C18PEG2000

B


Fold ICAM-2 protein silencing
(Normalized to 7C1)
**
6
4
2
0
Original 7C1
Screen 1 Winner
LNP1
BM1
LNP3
Screen 2


C
Diameter (nm)
Days after formulation
D
7C1
Fluorescence Intensity
pH
E
BM1
Fluorescence Intensity
pH
F
7C1
Untreated
No Inhibitor
+Genistein
+Chlopromazine
+7C1- si647
Alexa647
G
BM1
Untreated
No Inhibitor
+Genistein
+Chlopromazine
+BM1- si647
Alexa647
H
7C1
Barcode Counts (AU)
Heart ECs
Lung ECs
Marrow ECs
CD34+ HSPC
I
BM1
Barcode Counts (AU)
Heart ECs
Lung ECs
Marrow ECs
CD34+ HSPC
J
7C1
BM1
**
Normalized Nucleic Acid Delivery
(fold increase compared to 7C1)
Heart ECs
Lung ECs
Marrow ECs
CD34+ HSPC
K
sgICAM2:
5'-asasgsACGGACAGGCACCUACGGUUUUAGAgc
uagaaauagcAAGUUAAAAUAAGGCUAGUCCGUUA
UCAacuugaaaaaguggcaccgagucggugcusususu-3'
A, G, U, C: RNA nucleotide
a, g, u, c: 2'-O-Methyl nucleotide
s: phosphorothioate
L
CD34+ HSPC
P = 0.29
ICAM-2 Indel (%)
PBS
BM1
M
PBS
BM1 + siRNA
% initial Bodyweight
0hrs
24hrs
72hrs
N
PBS
BM1 + sgRNA
% initial Bodyweight
0hrs
24hrs
72hrs